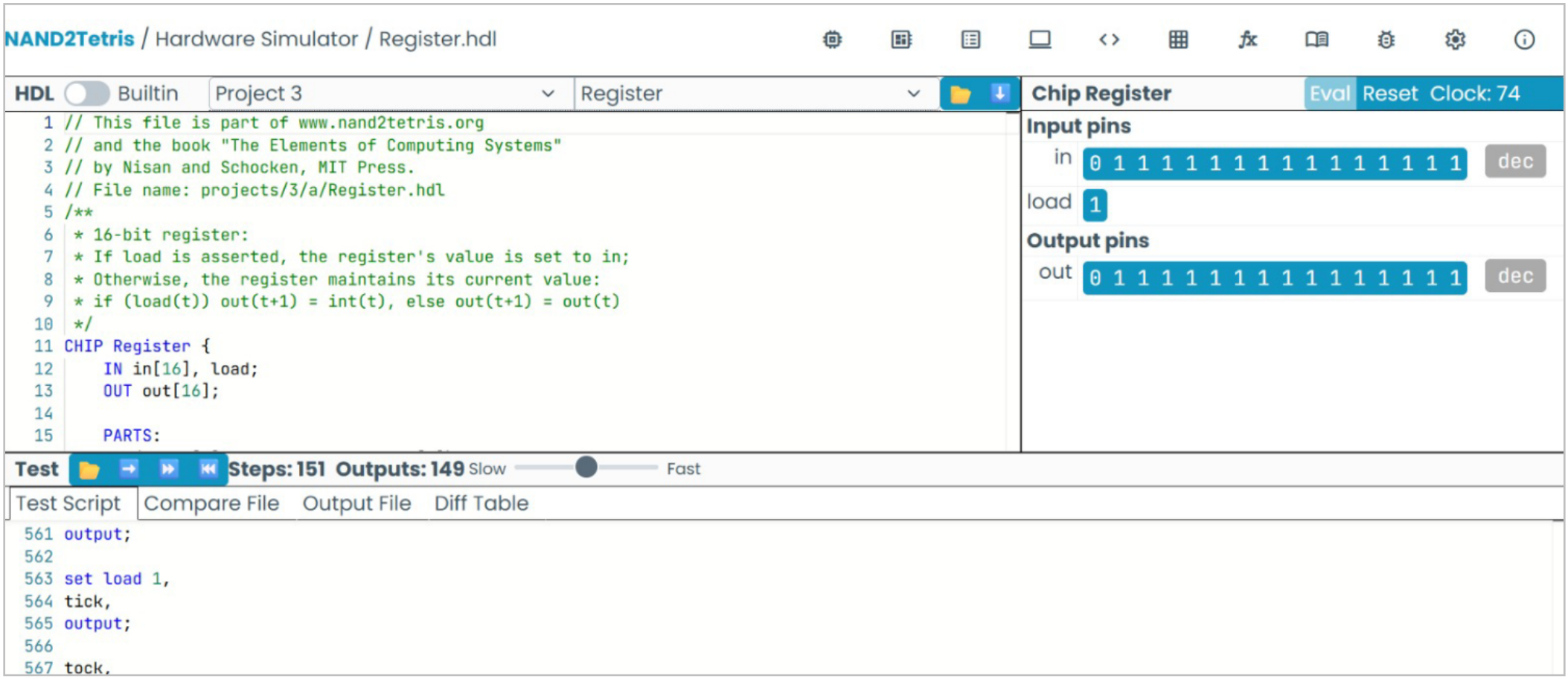
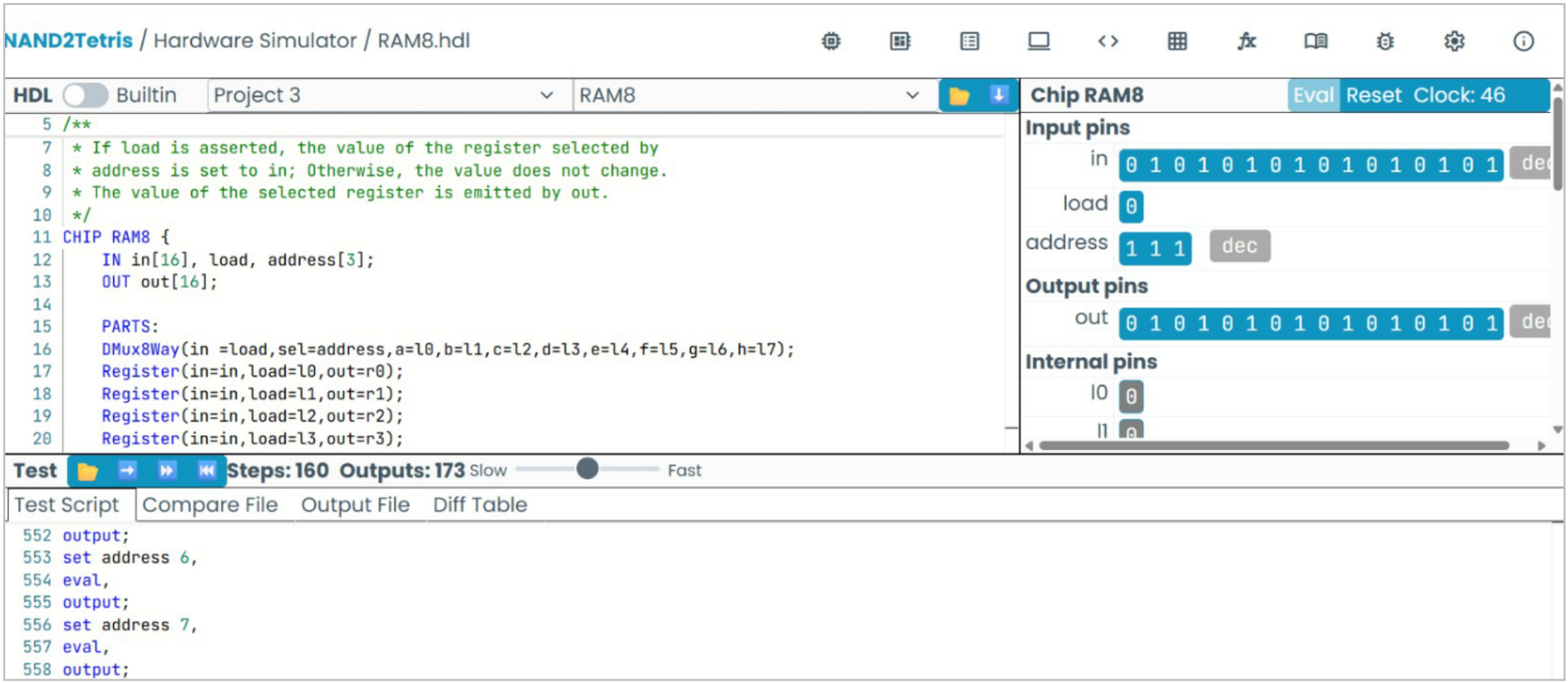
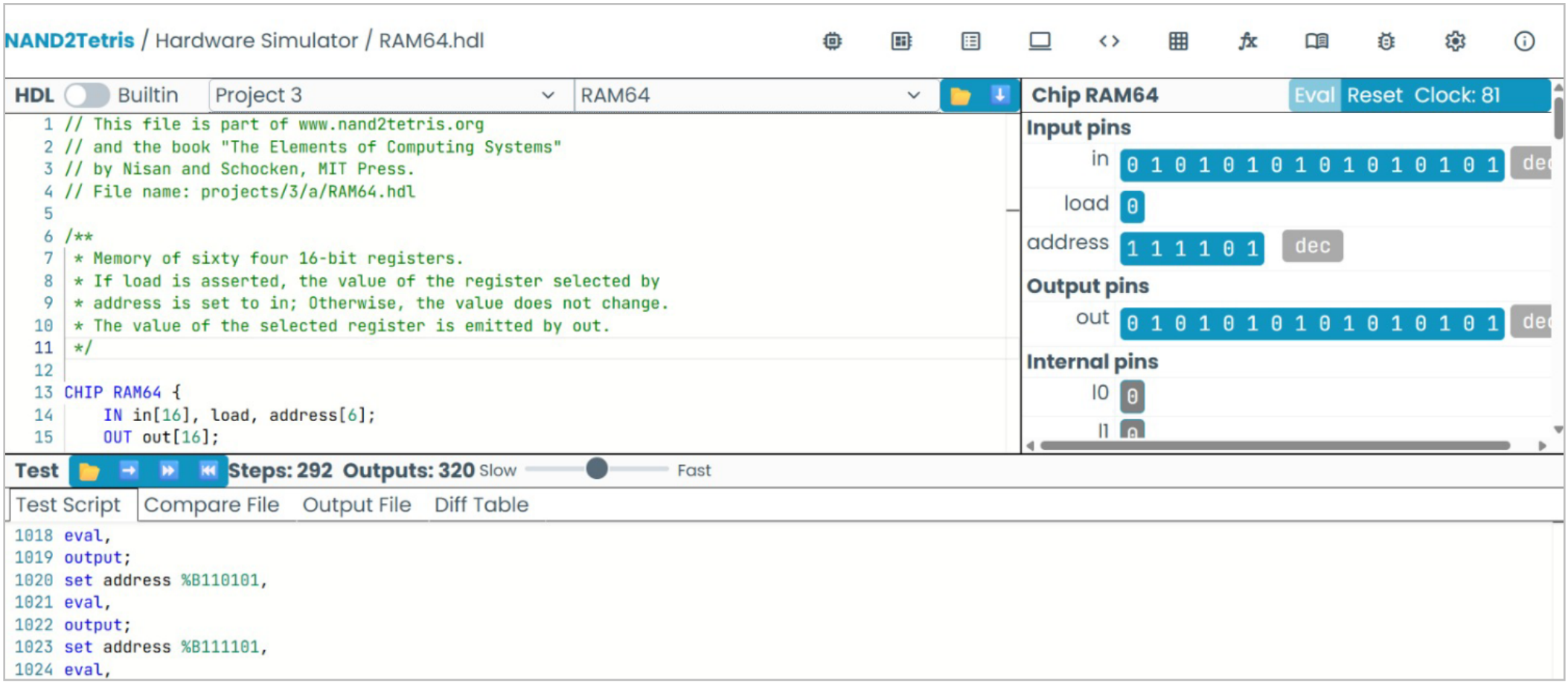
1. Register.hdl



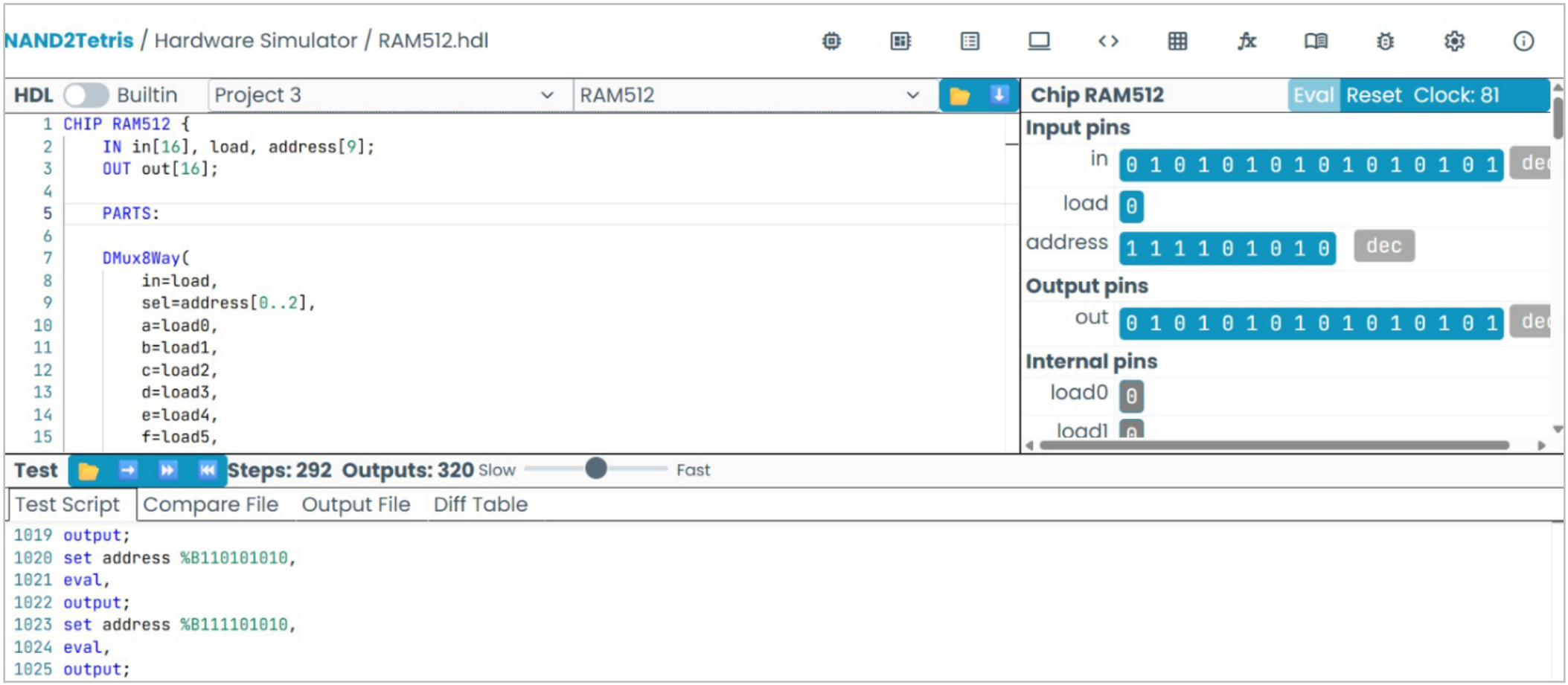
2. RAM8.hdl


NAND2Tetris / Hardware Simulator / RAM8.hdl
HDL
Builtin
Project 3
RAM8
5 /**
7 * If load is asserted, the value of the register selected by
8 * address is set to in; Otherwise, the value does not change.
9 * The value of the selected register is emitted by out.
10 */
11 CHIP RAM8 {
12 IN in[16], load, address[3];
13 OUT out[16];
15 PARTS:
16 DMux8Way(in =load,sel=address,a=l0,b=l1,c=l2,d=l3,e=l4,f=l5,g=l6,h=l7);
17 Register(in=in,load=l0,out=r0);
18 Register(in=in,load=l1,out=r1);
19 Register(in=in,load=l2,out=r2);
20 Register(in=in,load=l3,out=r3);
Chip RAM8
Eval
Reset
Clock: 46
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
load
0
address
1 1 1
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
Internal pins
l0
0
Test
Steps: 160 Outputs: 173
Slow
Fast
Test Script
Compare File
Output File
Diff Table
552 output;
553 set address 6,
554 eval,
555 output;
556 set address 7,
557 eval,
558 output;

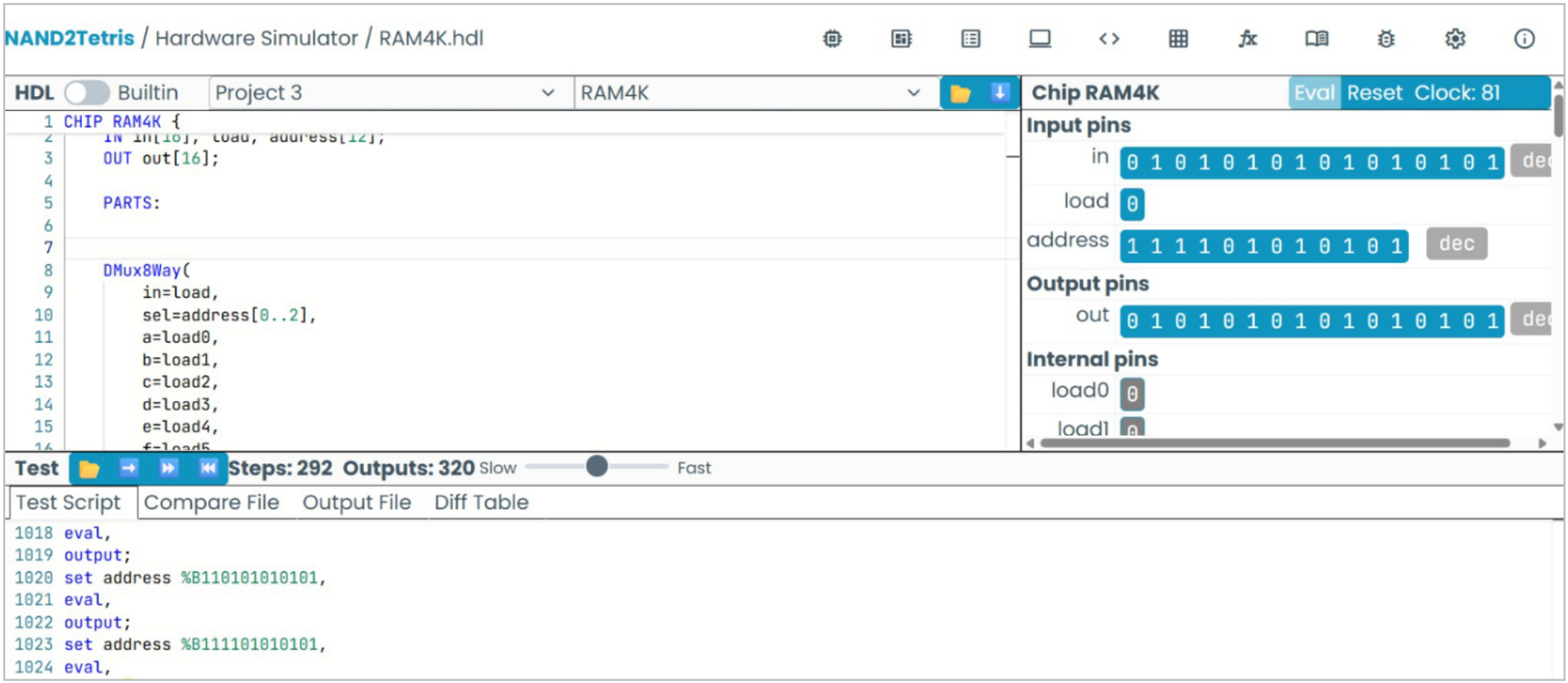
3. RAM64.hdl


NAND2Tetris / Hardware Simulator / RAM64.hdl
HDL
Builtin
Project 3
RAM64
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/3/a/RAM64.hdl
5
6 /**
7 * Memory of sixty four 16-bit registers.
8 * If load is asserted, the value of the register selected by
9 * address is set to in; Otherwise, the value does not change.
10 * The value of the selected register is emitted by out.
11 */
12
13 CHIP RAM64 {
14 IN in[16], load, address[6];
15 OUT out[16];
Chip RAM64
Eval
Reset
Clock: 81
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
load
0
address
1 1 1 1 0 1
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
Internal pins
l0
0
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Compare File
Output File
Diff Table
1018 eval,
1019 output;
1020 set address %B110101,
1021 eval,
1022 output;
1023 set address %B111101,
1024 eval,

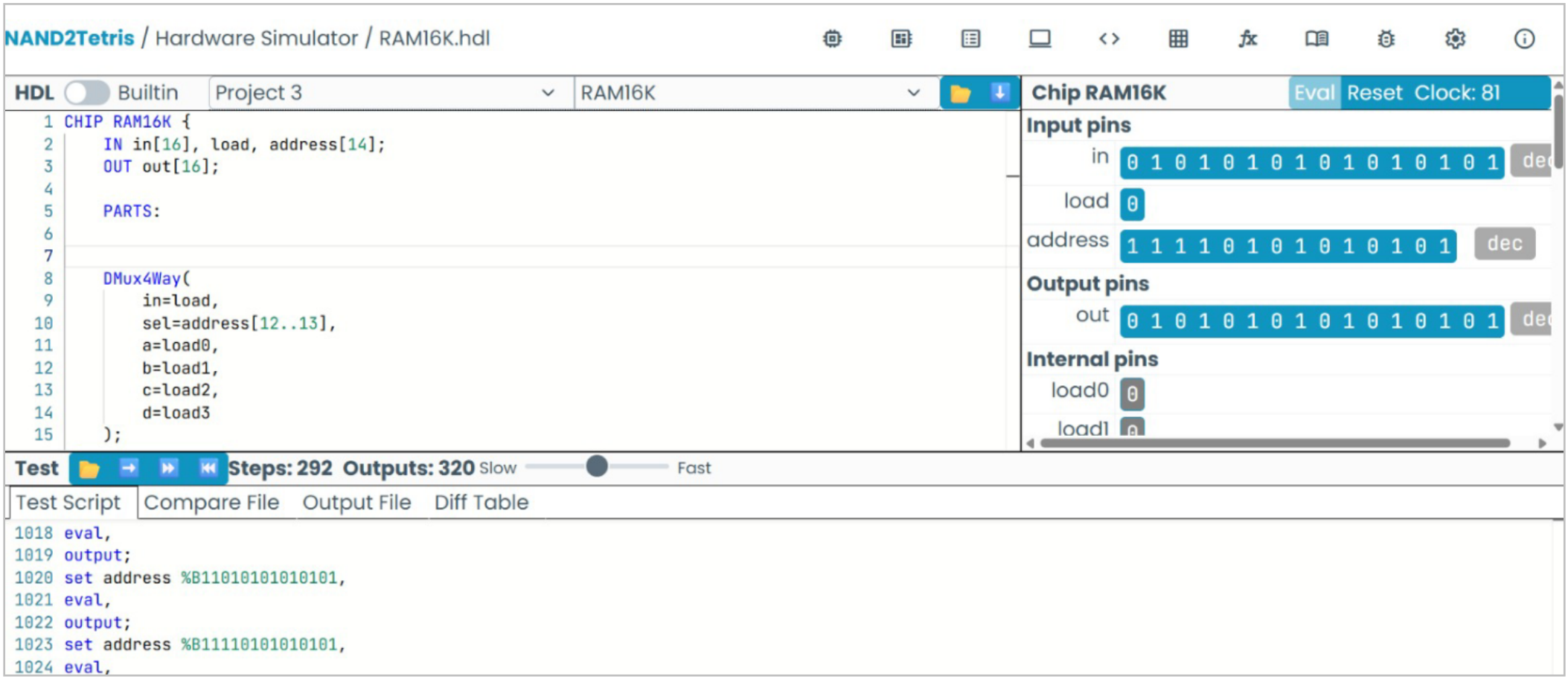
4. RAM512.hdl


NAND2Tetris / Hardware Simulator / RAM512.hdl
HDL
Builtin
Project 3
RAM512
1 CHIP RAM512 {
2 IN in[16], load, address[9];
3 OUT out[16];
4
5 PARTS:
6
7 DMux8Way(
8 in=load,
9 sel=address[0..2],
10 a=load0,
11 b=load1,
12 c=load2,
13 d=load3,
14 e=load4,
15 f=load5,
Chip RAM512
Eval
Reset
Clock: 81
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
load
0
address
1 1 1 1 0 1 0 1 0
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
Internal pins
load0
0
load1
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Compare File
Output File
Diff Table
1019 output;
1020 set address %B110101010,
1021 eval,
1022 output;
1023 set address %B111101010,
1024 eval,
1025 output;

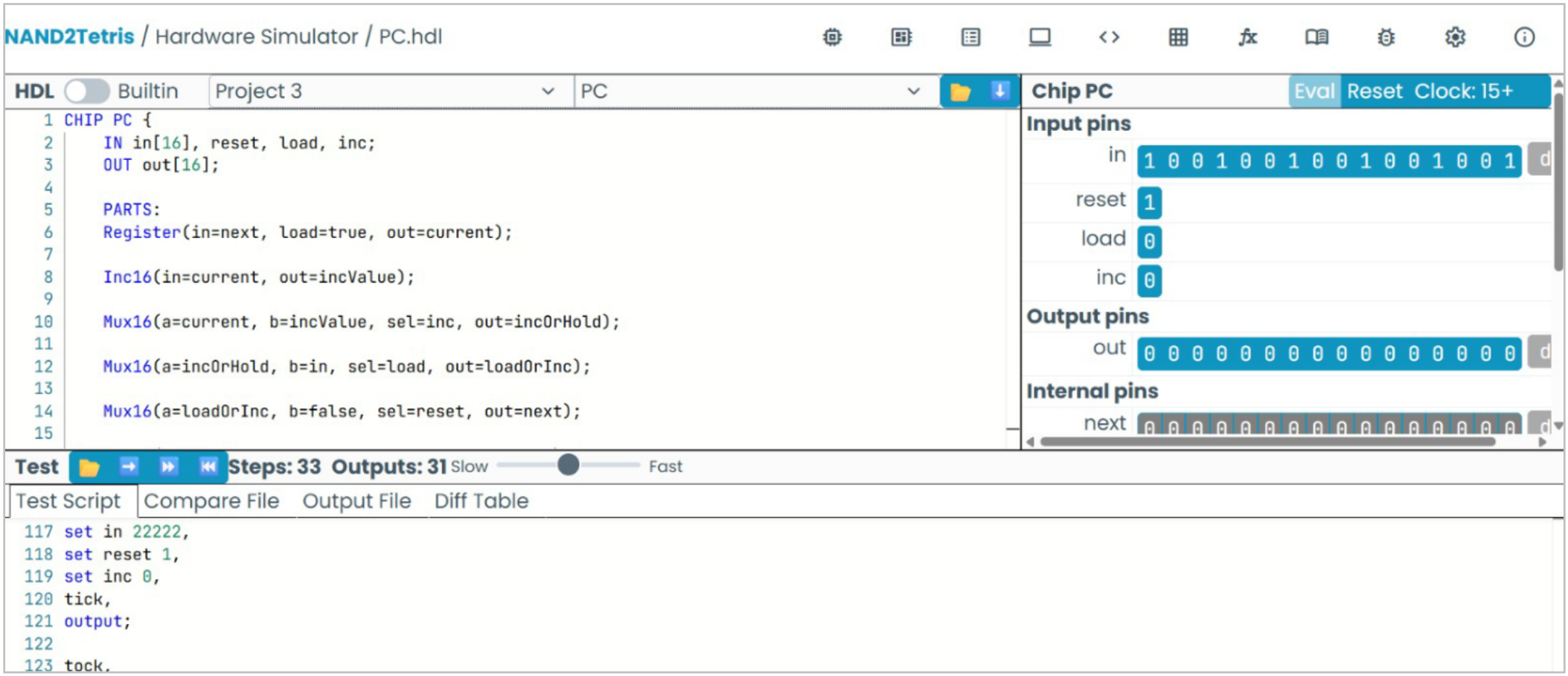
5. RAM4K.hdl


NAND2Tetris / Hardware Simulator / RAM4K.hdl
HDL
Builtin
Project 3
RAM4K
1 CHIP RAM4K {
3 OUT out[16];
5 PARTS:
8 DMux8Way(
9 in=load,
10 sel=address[0..2],
11 a=load0,
12 b=load1,
13 c=load2,
14 d=load3,
15 e=load4,
Chip RAM4K
Eval
Reset
Clock: 81
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
load
0
address
1 1 1 1 0 1 0 1 0 1 0 1 0 1 0 1
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
Internal pins
load0
0
load1
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Compare File
Output File
Diff Table
1018 eval,
1019 output;
1020 set address %B110101010101,
1021 eval,
1022 output;
1023 set address %B111101010101,
1024 eval,

6. RAM16K.hdl


NAND2Tetris / Hardware Simulator / RAM16K.hdl
HDL
Builtin
Project 3
RAM16K
1 CHIP RAM16K {
2 IN in[16], load, address[14];
3 OUT out[16];
4
5 PARTS:
6
7
8 DMux4Way(
9 in=load,
10 sel=address[12..13],
11 a=load0,
12 b=load1,
13 c=load2,
14 d=load3
15);
Chip RAM16K
Eval
Reset
Clock: 81
Input pins
in
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
load
0
address
1 1 1 1 0 1 0 1 0 1 0 1 0 1
dec
Output pins
out
0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1
Internal pins
load0
0
load1
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Compare File
Output File
Diff Table
1018 eval,
1019 output;
1020 set address %B11010101010101,
1021 eval,
1022 output;
1023 set address %B11110101010101,
1024 eval,

7. PC.hdl


NAND2Tetris / Hardware Simulator / PC.hdl
HDL
Builtin
Project 3
PC
1 CHIP PC {
2 IN in[16], reset, load, inc;
3 OUT out[16];
4
5 PARTS:
6 Register(in=next, load=true, out=current);
7
8 Inc16(in=current, out=incValue);
9
10 Mux16(a=current, b=incValue, sel=inc, out=incOrHold);
11
12 Mux16(a=incOrHold, b=in, sel=load, out=loadOrInc);
13
14 Mux16(a=loadOrInc, b=false, sel=reset, out=next);
15
Chip PC
Eval
Reset
Clock: 15+
Input pins
in 1 0 0 1 0 0 1 0 0 1 0 0 1 0 0 1
reset 1
load 0
inc 0
Output pins
out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Internal pins
next 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
Test
Steps: 33 Outputs: 31
Slow
Fast
Test Script
Compare File
Output File
Diff Table
117 set in 22222,
118 set reset 1,
119 set inc 0,
120 tick,
121 output;
122
123 tock,

8. Bit.hdl

```
// File name: projects/3/a/Bit.hdl
/**
 * 1-bit register:
 * If load is asserted, the register's value is set to in;
 * Otherwise, the register maintains its current value:
 * if (load(t)) out(t+1) = in(t), else out(t+1) = out(t)
 */
CHIP Bit {
    IN in, load;
    OUT out;

    PARTS:
    Mux(a=outLoop, b=in, sel=load, out=muxOut);
    DFF(in=muxOut, out=outLoop, out=out);
}
```

Input pins

in	1
load	1

Output pins

out	0

Internal pins

outLoop	0
muxOut	1

Test **Steps: 217 Outputs: 215** Slow Fast

Test Script | Compare File | Output File | Diff Table

```
set in 1,
set load 0,
tick,
output;

tock,
```